Measuring the Universe with Sound Waves

A From-Scratch Cosmology Simulation + Analysis Pipeline

Dennis Wu

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PHY 305 – Creative Project

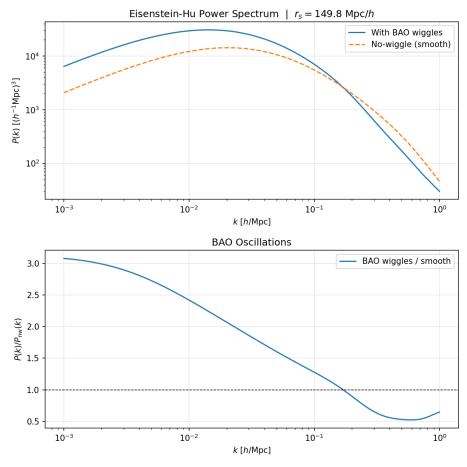
The big idea: a sound wave left a ruler in the sky

Before atoms formed ($t < 380{,}000$ yr): the universe was a hot photon–baryon plasma coupled by Thomson scattering.

Overdensities launched **spherical sound waves** at $c_s \approx c/\sqrt{3}$. When photons decoupled, the waves **froze in place** at a comoving radius $r_s \approx 150\ \mathrm{Mpc}$ — a fixed ruler in the matter distribution.

Today we see it as a **bump in** $\xi(r)$ near 100 h^{-1}Mpc and **wiggles in** $P(k)$.

The CMB pins r_s exactly, so BAO is a cosmic standard ruler — measuring its apparent size at any redshift gives $H(z)$, $d_A(z)$, and constraints on dark energy.



Linear $P(k)$ (top) and the wiggle ratio $P(k)/P_{\text{no-wiggle}}(k)$ (bottom). Those few-percent oscillations *are* BAO.

Why measuring BAO is hard

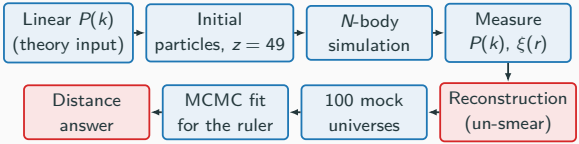
Problem 1 — gravity smears the ruler. Galaxies drift toward overdense regions, blurring the BAO bump by several h^{-1}Mpc.
Fix: reconstruction — estimate where galaxies came from and move them back.

Problem 2 — the wiggles ride on a huge smooth signal. Our simulation's finite resolution also distorts that smooth shape.
Fix: broadband marginalization — fit the smooth shape as a nuisance; only the oscillations constrain the ruler.

Problem 3 — one universe = one noisy measurement.
Fix: 100 fake "mock" universes to estimate error bars (a covariance matrix).

This project = a pipeline that simulates a piece of universe, measures BAO, fixes all three problems, and checks the answer.

The pipeline in one picture



What I simulated. A cubic box $L = 1.5\ \mathrm{Gpc}/h$ on a side with 128^3 dark-matter particles on a 256^3 force mesh, evolved from $z = 49$ to today in 50 time steps.

Every stage written from scratch (NumPy + SciPy) except the `pyrecon` reconstruction step (fair warning: I used an existing library for that).

How the N-body simulation works

Particle–mesh gravity

Computing gravity between every pair of particles is $\mathcal{O}(N^2)$ — way too slow for $128^3 \approx 2$ million particles.

Instead:

1. Smear each particle onto a grid (*cloud-in-cell*, like anti-aliasing).
2. Solve Poisson's equation on the grid in Fourier space: $\hat{\Phi}(\mathbf{k}) \propto \hat{\delta}(\mathbf{k})/k^2$. One FFT turns a PDE into a division.
3. Take gradient $\rightarrow$ force field $\rightarrow$ interpolate back to each particle.

Cost drops from $\mathcal{O}(N^2)$ to $\mathcal{O}(N_{\mathrm{mesh}}^3 \log N_{\mathrm{mesh}})$.

Leapfrog time integrator

A *symplectic* stepper — the same family used for planetary orbits, because it conserves energy over *billions* of steps where plain Runge–Kutta would drift.

Conceptually:

$$\texttt{kick} \rightarrow \texttt{drift} \rightarrow \texttt{kick}$$

(update velocity by half a step, update position by a full step, update velocity by another half step).

We step in the scale factor a, not time, so that each step spans a similar fraction of cosmic history.

From one simulation to error bars and a fit

How big are my error bars?

One $P(k)$ from one box is noisy. I generate **100 fast "mock" universes** using the lognormal trick (draw a Gaussian field, exponentiate it — cheap and keeps densities positive), measure $P(k)$ on each, and take the scatter:

$$C_{ij} = \mathrm{cov}\big(P_i,\, P_j\big) \quad \text{over 100 mocks.}$$

That's my covariance matrix — the error bars plus all their correlations.

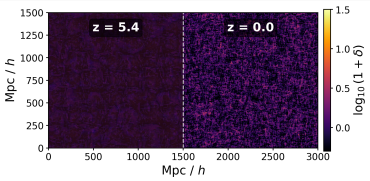
What am I fitting?

A template: a smooth background polynomial plus the BAO wiggle pattern, with two knobs:

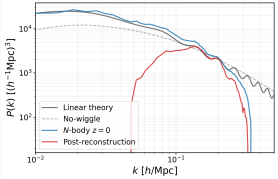
- α = ruler stretch factor ($\alpha = 1$ means "ruler matches the expected length").
- Σ_{nl} = how much gravity has smeared the wiggles (in h^{-1}Mpc).

The background polynomial coefficients are *analytically integrated out* (a linear-algebra trick). Only α and Σ_{nl} are sampled by MCMC (Metropolis–Hastings, 20k steps).

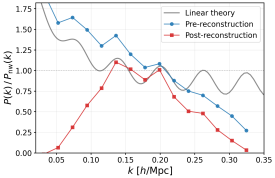
What came out of the simulation



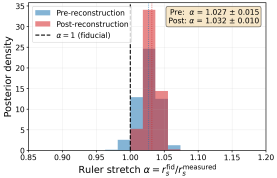
(a) Cosmic web grows.
Gravity turns a nearly uniform start ($z = 5.4$) into filaments, walls, and voids today ($z = 0$).



(b) Power spectrum $P(k)$.
How lumpy the universe is at each scale. Blue = simulation; gray = theory.

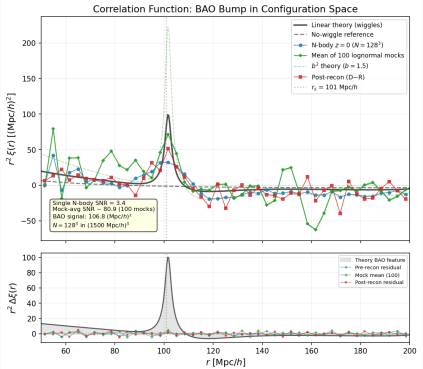


(c) BAO wiggle ratio $P/P_{\rm nw}$.
Divide out the smooth shape to isolate the few-percent acoustic wiggles.



(d) Ruler measurement.
MCMC posterior on α. Narrower after recon (red) $\Rightarrow$ more precise distance.

What came out: the ruler measurement



The BAO bump in the correlation function $\xi(r)$ at $r \approx 101\ h^{-1}$Mpc — there it is.

Fit results

	Before recon	After recon
Ruler α	1.027 ± 0.015	1.032 ± 0.010
Smearing $\Sigma_{\rm nl}$	$4.8\ h^{-1}$Mpc	$3.0\ h^{-1}$Mpc

- The recovered ruler is within 3% of the expected length ($r_s = 100.9\ h^{-1}$Mpc).
- Reconstruction tightens the error bar by 1.6× and reduces the smearing — exactly what it's designed to do.
- In the correlation function I see the bump with signal-to-noise ≈ 3.4 in one box, and ≈ 80 after averaging 100 mocks.

How I know the code is right

- **Null test:** feed in uniform-random particles $\Rightarrow$ the estimator returns $P(k) \approx 0$ after shot-noise subtraction. It does.
- **Initial-conditions round-trip:** at $z = 49$ the particles should give $P(k) = D^2(49)\, P_{\rm lin}(k)$ with $D(49) \approx 0.02$. They do.
- **Linear regime:** at large scales ($k \lesssim 0.05\ h/\mathrm{Mpc}$) my simulation matches linear theory; at smaller scales it (correctly) exceeds it.
- **Broadband bias check:** without marginalization, $\alpha \approx 1.19$ (19% off). With marginalization, $\alpha \to 1$ on the *same* data.
- **Mock sanity:** 100 lognormal mocks reproduce the input $P(k)$ within cosmic variance.
- **Unit tests:** `pytest tests/` covers every module.

Takeaways and future directions

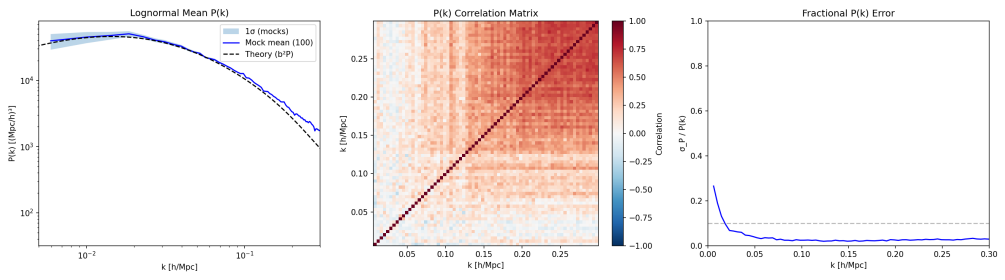
What this project shows

- A from-scratch N-body + MCMC pipeline can recover a cosmic ruler to the $\sim$1% level using only $(1.5\ \mathrm{Gpc}/h)^3$ of simulated volume.
- Reconstruction visibly sharpens the BAO bump and shrinks the error bar by 1.6$\times$ — concrete evidence that un-smearing bulk flows works.
- Marginalizing the smooth background is essential; skipping it produces a 19% bias on the ruler.

Where this could go next

- Bigger / higher-resolution box (tree-PM gravity) to resolve the wiggles directly.
- Redshift-space effects; survey masks; multiple tracers.
- Fit the ruler separately along vs. across the line-of-sight ($\alpha_\parallel$, $\alpha_\perp$).

Backup: what the covariance looks like



Left: error bar on $P(k)$ per bin. Middle: correlation matrix — off-diagonal leakage is small. Right: mean mock $P(k)$ matches the theory input.

Backup: why we marginalize the background

The fitting template splits into two pieces:

$$P_{\mathrm{model}}(k) \;=\; \underbrace{O_{\mathrm{wig}}(k;\,\alpha,\,\Sigma_{\mathrm{nl}})}_{\text{BAO wiggles, nonlinear}} \;+\; \underbrace{a_{-1}/k + a_0 + a_1 k}_{\text{smooth background}}.$$

The background coefficients $\{a_{-1}, a_0, a_1\}$ are *nuisance parameters* — we don't care about them, but resolution effects in the simulation do affect them.

Because they enter *linearly*, they can be integrated out of the Gaussian likelihood in closed form. The MCMC then only has to explore two dimensions $(\alpha, \Sigma_{\mathrm{nl}})$ instead of five — faster to converge, and the ruler is now insulated from any smooth shape distortion.

Consequence: without marginalization, a bent broadband drags α to $\sim$1.19. With marginalization, $\alpha \to 1$. Same data, same simulation — the only change is how we fit.